

Sagnik Datta

[Phone](#) | [Mail](#) | [LinkedIn](#) | [GitHub](#)

Education

Vellore Institute of Technology

Bachelor of Technology in Computer Science & Engineering (IoT Specialization)

Vellore, India

2024 – Present

G D Goenka Public School

Senior Secondary School Certificate (CBSE Class XII) – Computer Science

Dakshineswar, India

2022 – 2024

Don Bosco School

Indian Certificate of Secondary Education (ICSE Class X) – Science

Liluah, India

2011 – 2022

Technical Skills

Programming Languages: Python, C++, TypeScript, JavaScript (ES6+), SQL (Oracle/PostgreSQL), C, HTML5, CSS3

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, FastAPI, Prisma ORM, WebSockets, Tailwind CSS, Streamlit

AI, ML & Data Systems: Local LLMs (LLaMA 3, Ollama), Retrieval-Augmented Generation (RAG), Vector Embeddings, NLP, Prompt Engineering

Cloud, DevOps & Databases: Oracle Cloud Infrastructure (OCI), Oracle Database, Docker, Kubernetes, CI/CD Pipelines, PostgreSQL, Git, Linux/Bash

Engineering Methodologies: Agile, Scrum, Test-Driven Development (TDD), Unit Testing, Root-Cause Debugging, System Design, REST APIs, UML

Work Experience

Skyscanner

Remote / Virtual

Front-End Software Engineering Virtual Experience (Forage)

2026

- Engineered modular, responsive web user interfaces within an Agile sprint lifecycle using Skyscanner's Backpack React UI library.
- Refactored dynamic client state management pipelines, slashing component re-renders by 20% and improving page load speed by 15%.
- Conducted automated unit testing and root-cause debugging across multiple viewports, achieving 99.8% client-side application uptime.

Electronic Arts (EA)

Remote / Virtual

Software Engineering Virtual Experience (Forage)

2025

- Architected object-oriented backend class models and state-rollback algorithms for a proposed "Rewind Dynasty Mode" in C++.
- Formulated detailed UML class and sequence diagrams to align microservices specifications across the software development lifecycle (SDLC).
- Profiled system execution and resolved concurrency race conditions using automated unit tests, enhancing runtime stability by 25%.

Projects

Multi-User Real-Time Video Binging Platform | [Live Demo](#) | [GitHub](#)

- Architected a synchronized multi-user video streaming platform featuring room management and sub-50ms playback sync.
- Engineered peer-to-peer WebRTC connections and WebSocket channels for synchronized media playback and low-latency group chat.

SaaS Event & Ticketing Platform | [GitHub](#)

- Architected a containerized multi-tenant microservices backend handling 500+ concurrent real-time transactions with zero data loss.
- Integrated persistent WebSocket channels to ensure sub-100ms real-time seat inventory synchronization across distributed clients.

Real-Time Crypto Arbitrage Detection Engine | [Frontend](#) | [Backend](#) | [GitHub](#)

- Architected end-to-end system design diagrams, deployment runbooks, and project setup pipelines for seamless cross-service integration.
- Developed interactive front-end page interconnections, custom terminal-styled UI components, and client-side view routing.

Noto.ai – Offline Document Intelligence Assistant | [GitHub](#)

- Engineered a 100% offline RAG document assistant utilizing local LLaMA 3 inference via Ollama for zero-data-leakage privacy.
- Developed chunked text parsing and contextual vector search indexing with a modular Kivy cross-platform interface.

Interactive Retro Windows 98 Portfolio | [Live Demo](#) | [GitHub](#)

- Engineered an authentic Windows 98 desktop simulation featuring draggable windows, active taskbar, and virtual filesystem.
- Implemented modular window lifecycle state management and retro audio synthesis routines with zero third-party dependencies.

Honors, Achievements & Certifications

- Honors** – Two-Time Finalist, ACM-VIT Reverse Coding Competition (Consistently ranked in Top 20% / Top 50 of 180+ teams, 2025 & 2026)
- Certifications** – Oracle Certified Associate (OCI & Database), [Python: Data Structures \(Great Learning\)](#), [Generative AI for Beginners \(Great Learning\)](#)